MANGROVES MONITORING WITH ALPHAEARTH FOUNDATIONS AND SELF-ATTENTION

Abstract—**Mangrove forests are ecologically important but hard to map at scale because of tides, cloud cover, and the difficulty of telling mangroves apart from other coastal vegetation. Existing products like the Global Mangrove Watch (GMW) only give a binary yes/no label, which misses important details about how dense the mangrove cover actually is. We propose a pipeline that uses AlphaEarth Foundations–64-dimensional per-pixel embeddings that combine optical, SAR, LiDAR, and thermal satellite data from 2017 to 2024– to turn GMW labels into six coverage classes (0%, 1-20%, 21-40%, 41-60%, 61-80%, 81-100%). A Vision Transformer (ViT) trained with self-attention learns the spatial patterns in each 244×244 image chip. Grad-CAM maps show that the model focuses on the right ecological features. The pipeline works without any external ground truth labels and can be applied anywhere in the world.**

I. INTRODUCTION

Mangrove forests cover barely 0.1% of the Earth's land surface yet store four to ten times more carbon per unit area than most other forests (Kauffman et al. [1]).

Beyond carbon, they protect coastlines, support biodiversity and provide services whose value depends on how accurately their extent is mapped (Cárdenas et al. [2]; Farzanmanesh et al. [3]).

In "Blue Carbon" markets, boundary errors lead directly to errors in carbon stock estimates, which reduces the credibility of issued credits (Merchant et al. [4]).

The Global Mangrove Watch v3 (Bunting et al. [5]) is the most widely used global mangrove map, but it has known problems : boundaries are imprecise in tidal zones, narrow strips of fringing mangroves are often missed and the map only gives binary presence/absence labels at 100 m resolution.

Traditional classifiers that use single-date optical or radar images struggle even more because tidal flooding, cloud cover and spectral similarity between mangroves and dense forests make reliable classification very difficult (Tran et al. [6]; Giri et al. [7]). Broader reviews of satellite-based mangrove monitoring confirm these limitations across several decades of research (Purnamasayangsukasih et al. [8]; Wang et al. [9]; Maurya et al. [10]).

These limitations are clearly visible when comparing the GMW map with available reference imagery.

Fig. 1 shows Pulau Ubin, a mangrove-rich island northeast of Singapore, where the National Parks Board (NParks) mask and the Sentinel-2 image reveal fringing mangrove strips and tidal zones that GMW either misclassifies or misses entirely.

Mangroves are shown in darker green in the NParks mask (Fig. 1b) and in white in the GMW map (Fig. 1c).

This gap motivated our approach : instead of relying on external ground truth, our pipeline learns to improve GMW labels directly from the information encoded in AlphaEarth embeddings.

We address these limitations by using AlphaEarth Foundations (Tran et al. [11]), a model from Google DeepMind that turns eight years of multi-modal satellite data into dense 64-dimensional vectors for every pixel on Earth. Our contributions are: (i) a coverage ratio formula that converts binary GMW polygons into six graduated classes using direct spatial overlap, including a principled way to generate non-mangrove samples; (ii) a global dataset that is balanced across geographic regions and (iii) a ViT trained on these embeddings that goes beyond binary mapping to produce fine-grained coverage predictions, verified through Grad-CAM visualizations.

II. METHODOLOGY

A. AlphaEarth Foundations

AlphaEarth Foundations (Tran et al. [11]) turns all available satellite observations for every pixel on Earth–optical, SAR, LiDAR, and thermal– into a single 64-dimensional vector per pixel, per year.

Data from 2017 to 2024 are combined into one representation, which smooths out noise from clouds, tides, and seasonal changes.

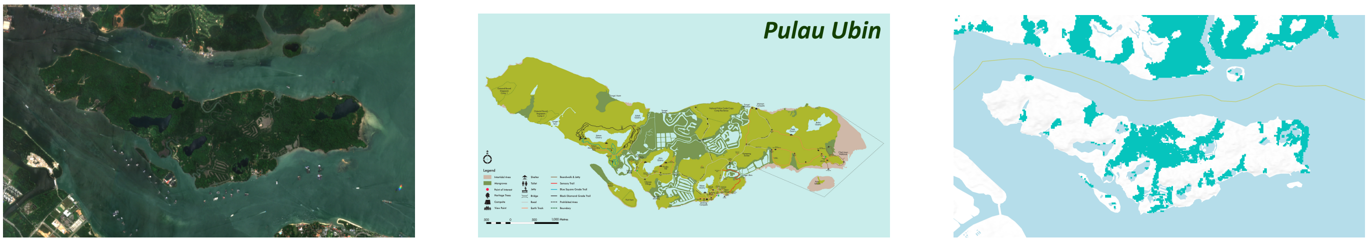
Each vector also captures what is happening within a 640 m radius around the pixel, so that even a single vector already encodes landscape context such as proximity to water, slope, and vegetation type (Fig. 2).

In our pipeline, embeddings are downloaded from Google Earth Engine as 244×244×64 tensors, using 20 parallel download threads. AlphaEarth embeddings have also been used for large-scale geospatial mapping tasks beyond mangroves, showing strong generalisation from sparse labels (Houriez et al. [12]).

B. Coverage Label Generation

One of the main contributions of this work is turning binary GMW polygon masks into a six-class coverage problem by computing the actual fraction of mangrove area within each image chip.

Let $\mathcal{G} = \{\mathcal{P}_1, \ldots, \mathcal{P}_N\}$ be the set of GMW v3 (2020) polygons in projected coordinates (EPSG:3857). For each polygon $\mathcal{P}_i$, we place a square chip B_i of side $s = Pr$

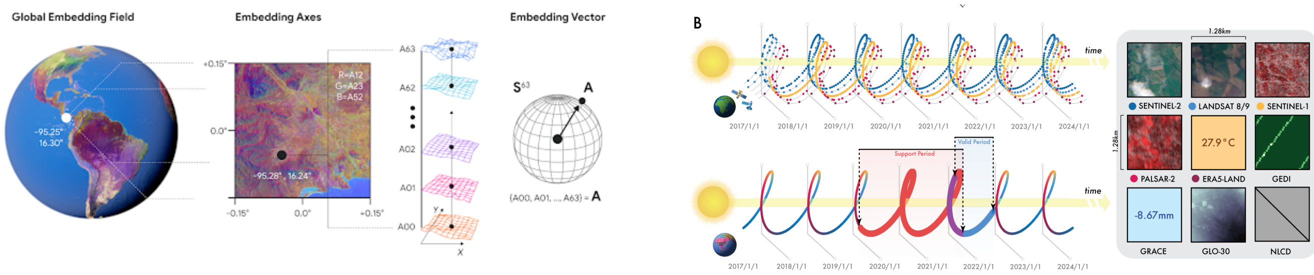


(a) Sentinel-2 image.

(b) NParks mask.

(c) NParks mask and GMW extent (Bunting et al. [5]).

Fig. 1. Pulau Ubin (Singapore): the Sentinel-2 image and the NParks mangrove mask reveal clear gaps in the GMW classification, particularly along tidal channels and narrow fringing strips. This motivated the label refinement approach presented in this paper.



(a) AlphaEarth Foundations architecture.

(b) Spatial context: each pixel captures a 640 m radius.

Fig. 2. AlphaEarth Foundations (Tran et al. [11]): (a) multiple satellite data sources are combined into one 64-dimensional vector per pixel; (b) each vector encodes information from within a 640 m radius, giving it built-in landscape context.

centered on the polygon's centroid $\mathbf{c}_i$, where $P = 244$ pixels and $r = 10$ m/pixel:

$$s = Pr = 2440 \text{ m}, \qquad A_{\text{chip}} = s^2. \tag{1}$$

The mangrove coverage ratio for chip i is:

$$\rho_i = \frac{1}{A_{\text{chip}}} \sum_{\{k : \mathcal{P}_k \cap B_i \neq \emptyset\}} \text{Area}\big(\mathcal{P}_k \cap B_i\big), \tag{2}$$

where the sum runs over all GMW polygons that overlap chip B_i.

To find these polygons quickly without checking all one million GMW polygons one by one, we use a spatial index–a tree structure that groups polygons by their location on the map, so that only nearby polygons are checked for each chip. The coverage class is then:

$$y_i = f(\rho_i) = \begin{cases} 0 & \text{if } \rho_i = 0 \\ \lceil 5\,\rho_i \rceil & \text{if } \rho_i > 0 \end{cases} \tag{3}$$

which maps the ratio to six classes: 0%, 1–20%, 21–40%, 41–60%, 61–80%, and 81–100%.

Category 0 (no mangrove). Any chip placed on a GMW polygon centroid will always contain at least some mangrove area, so $\rho_i > 0$ for all i, and class $y = 0$ can never appear naturally from GMW centroids.

We therefore generate no-mangrove samples separately : we draw random points within the mangrove belt ($|\phi| \leq 30°$ latitude), compute ρ for each using the same spatial index and keep only points where $\rho = 0$.

To make sure these non-mangrove samples are spread across the globe, this process is done independently for four geographic regions (Americas, Africa, Asia, Pacific), producing around 10 000 confirmed non-mangrove samples.

Geographic balance. The GMW shapefile has over one million polygons, but most of them are in Southeast Asia.

If we just sampled randomly from the full set, the training data would be heavily biased toward that region.

To fix this, we process up to $M = 50\,000$ polygons per geographic region and for each coverage class we sample proportionally across all four regions–so the model learns from mangrove ecosystems all over the world.

C. Vision Transformer Architecture

The input chip $\mathbf{X} \in \mathbb{R}^{P \times P \times D}$ (with $D = 64$ AlphaEarth channels) is cut into $n = (P/p)^2$ non-overlapping patches of size $p \times p$ pixels (Fig. 3). Each patch is turned into a vector of size d by a learned linear layer:

$$\mathbf{e}_j = \text{Flatten}(\mathbf{x}_j)\, W_{\text{proj}}, \quad j = 1, \ldots, n. \tag{4}$$

where $\mathbf{x}_j$ is the raw pixel block for patch j (a small cube of $p \times p \times 64$ numbers). $\text{Flatten}(\mathbf{x}_j)$ simply unrolls this cube into one long list of numbers.

W_{proj} is a learned weight matrix that compresses this long list into a shorter vector of size d–it is learned during training so that important information is kept. The result $\mathbf{e}_j$ is the patch embedding : a compact, fixed-size representation of what patch j looks like.

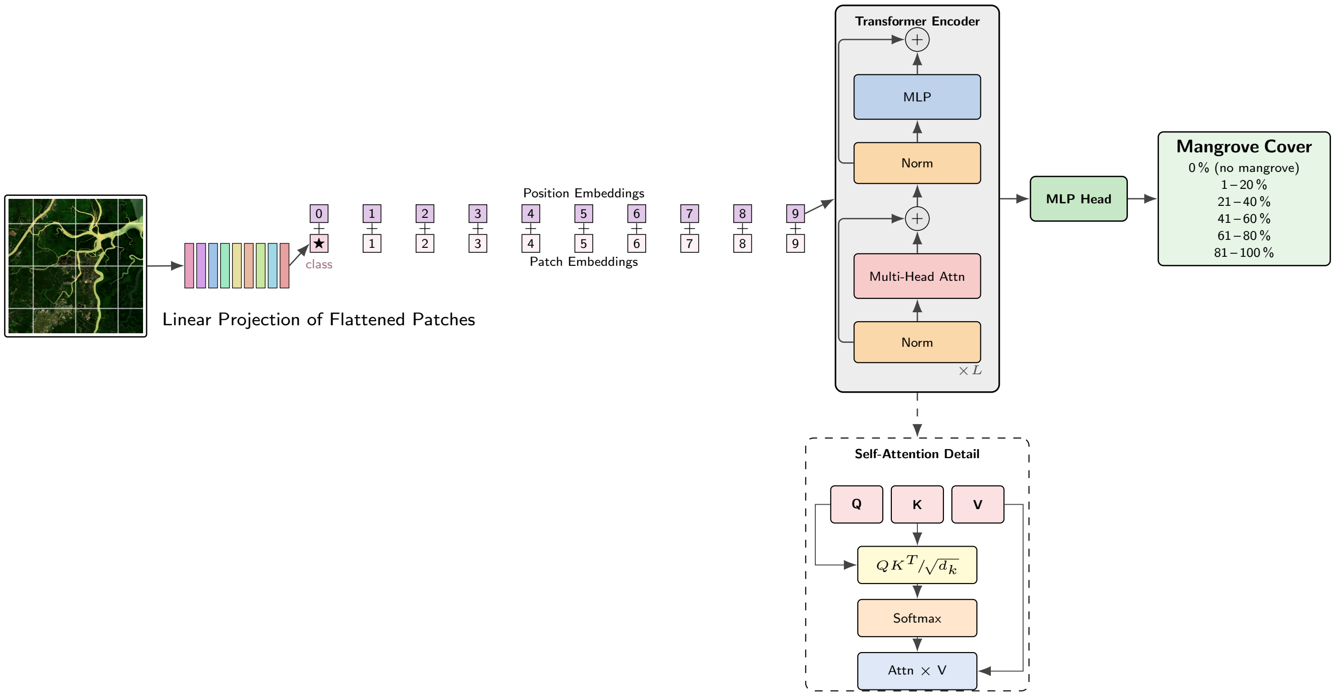


Fig. 3. Vision Transformer (ViT) architecture. The 244×244×64 input is split into 16×16 patches, projected into vectors, and passed through L encoder layers with multi-head self-attention and MLP blocks. A special summary token ([CLS]) collects information from all patches and feeds the final classification head, which predicts one of six coverage classes.

We then add a special extra token called [CLS] (short for "classification") : unlike the patch tokens, it does not represent any part of the image. Its job is to collect and summarize information from all the other patches through the attention layers.

At the end, only the [CLS] output is used to make the final class prediction. Position information is added to all tokens before they enter the encoder:

$$\mathbf{z}_j^{(0)} = \mathbf{e}_j + \mathbf{p}_j, \quad j = 0, \ldots, n. \tag{5}$$

where $\mathbf{p}_j$ is a learned *positional encoding* : a vector that encodes where patch j sits in the grid (top-left, center, bottom-right, etc.).

$\mathbf{z}_j^{(0)}$ is then the starting representation of token j–patch content plus position– that is fed into the first encoder layer.

Each of the L encoder layers applies multi-head self-attention (MHSA) with H heads. For a single head with size $d_k = d/H$:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{6}$$

where Q, K, and V are three different linear projections of the input. After the last encoder layer, the [CLS] token feeds a two-layer prediction head that outputs scores over the six coverage classes.

Self-attention lets every patch look at every other patch in the chip at the same time, which matters a lot for mangroves: ecosystem boundaries follow gradients (tidal channels, transitions between water and vegetation) that span many patches and are missed by convolutional networks with fixed local windows. Since the 64 AlphaEarth channels already carry multi-modal and multi-year information, the transformer only needs to learn how the patches relate to each other—not low-level visual features from scratch.

D. Complete Pipeline

The full pipeline (Fig. 4) runs in six steps:

1) **Coverage labels:** Eq. (2)–(3) assign each GMW centroid a class; non-mangrove samples ($y = 0$) are generated by geographic random sampling with a spatial filter.
2) **Embedding download:** AlphaEarth 244×244×64 tensors are retrieved from GEE using 20 parallel threads.
3) **Dataset:** Chips are balanced by coverage class and by geographic region (Americas, Africa, Asia, Pacific) to ensure global diversity.
4) **Training:** The ViT is trained with Adam, learning rate decay, cross-entropy loss, and early stopping based on validation loss via PyTorch Lightning.
5) **Inference:** The trained model predicts one of six coverage classes for each new chip.
6) **Interpretability:** Grad-CAM maps check that predictions are based on the right ecological features.

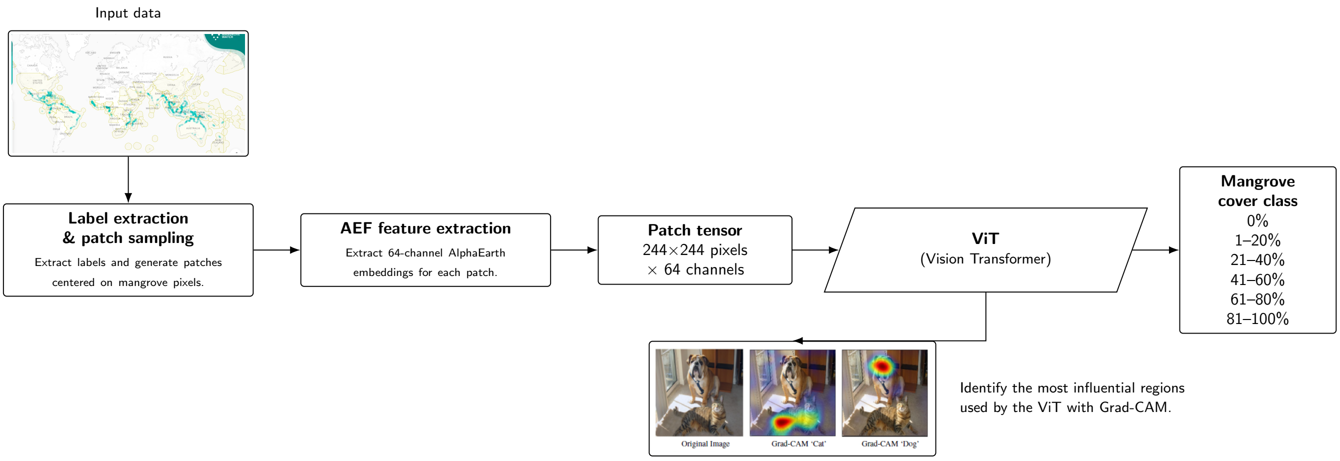


Fig. 4. Complete pipeline: GMW polygon labels are converted into six coverage classes using Eq. (2)–(3), AlphaEarth embeddings are downloaded from GEE, and a ViT is trained to predict the coverage class. Grad-CAM maps verify that the model focuses on the right ecological features.

III. Model Design

A. Baselines

To check how much self-attention actually helps, we compare the ViT against four simpler models trained on the same embedding tensors. Two classic machine learning baselines– Random Forest (RF) and XGBoost– take a simple 128-dimensional summary of each chip : the average and the standard deviation of each of the 64 embedding channels.

Two deep learning baselines–a Fully Connected Network (FCN) and a Convolutional Neural Network (CNN)– receive the full 244×244×64 tensor but can only look at a fixed local area at a time, which limits their ability to pick up on long-range spatial patterns. CNN-based models have already shown good results for mangrove mapping (Wan et al. [13]; Guo et al. [14]), but their fixed local view remains a key limitation when boundaries span large spatial scales.

All models are trained and tested on the same splits.

B. Training Details

The ViT uses patch size $p = 16$, embedding size $d = 256$, $L = 6$ encoder layers, $H = 8$ attention heads, and dropout 0.1.

We train with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, initial learning rate 10^{-3}), step-based learning rate decay, batch size 16, and automatic mixed precision on a single GPU, using PyTorch Lightning.

The baseline deep learning models use the same optimizer and run for 20 epochs.

C. Grad-CAM Interpretability

Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al. [15]) shows which parts of a chip drive the model's prediction. For a predicted class c, the heatmap is built in two steps. First, compute how much each feature map channel contributed to the prediction:

$$\alpha_k^c = \frac{1}{Z}\sum_{i,j}\frac{\partial S^c}{\partial A_{ij}^k}, \tag{7}$$

where S^c is the class score before the final softmax, A^k is the k-th feature map, and Z is the total number of positions. Then, combine the channels into a single heatmap:

$$L^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c\, A^k\right). \tag{8}$$

For our ViT, Grad-CAM is applied to the patch outputs of the last encoder block.

By looking at these maps across all six classes, we can check that the model is working correctly: high-coverage chips should highlight dense canopy areas, mid-coverage chips should focus on water–vegetation edges, and class-0 chips should point to open water or land. This step also helps catch failure cases, for example when the model confuses mangroves with dense inland forest, which can look similar in some regions (Giri et al. [7]).

IV. Discussion

A. Why Foundation Embeddings for Mangroves?

Traditional classifiers pick SAR images and some spectral bands from one or two sensors on a single date, usually chosen via specific algorithms to only use the most important ones. Additionally, they often rely on external ground truth labels that are expensive to collect and may not be available everywhere. Finally, they treat each pixel independently, without learning how different spatial areas relate to each other.

AlphaEarth solves all three problems at once : it combines optical, SAR, LiDAR, and thermal data; it uses eight years of observations instead of one date and its 640 m spatial context already encodes landscape-level features in each vector.

This is especially useful for mangroves, where a single satellite pass during high tide or a cloudy season can give a completely wrong picture (Giri et al. [7]).

Even a simple model trained on AlphaEarth embeddings should outperform a carefully tuned classifier using raw single-source imagery–our ViT goes further by learning how different spatial areas in the chip relate to each other.

B. Six Classes vs. Binary Maps

The six-class formula in Eq. (3) is a real step beyond binary GMW maps.

First, the 20% intervals between classes capture ecosystem gradients (sparse edge mangroves vs. dense interior forest) that matter for carbon estimates : biomass is not the same across every pixel and a coverage ratio gives a much better proxy for above-ground carbon than a simple yes/no label.

Second, class 0 is built from confirmed non-mangrove points found by spatial filtering–not just assumed from GMW absence–which cuts down on false positives in mixed coastal areas.

Together, the six-class labels and global sampling let the model work across very different mangrove types, from Caribbean red mangroves to Southeast Asian Rhizophora forests, without needing to be retrained for each region.

C. No External Ground Truth

Our pipeline is fully self-contained : training labels come from GMW and model quality is checked through Grad-CAM and by comparing predictions with the NParks mangrove mask for Pulau Ubin.

We do not use the CoastTrain dataset (Bunting et al. [5]) as ground truth because it was built using the same GMW classification process; using it would create a circular evaluation.

The NParks mask is also not used for training, only as a visual reference, since it dates from several years ago and may not reflect the current state of the mangroves.

Instead, our model improves on GMW by using the richer information already encoded in AlphaEarth vectors.

D. Limitations and Future Work

AlphaEarth averages data over eight years, which reduces noise from tides but may also hide short-term inundation patterns that help distinguish mangrove species (Avicennia, Rhizophora, Sonneratia).

Our pipeline maps coverage, not species composition, which limits its use for biodiversity studies.

Species-level mapping would require hyperspectral data or field-collected spectral measurements.

We have ordered 1 m resolution SAR imagery from TeLEOS-2 and optical imagery from DS-EO for Pulau Ubin.

These images will be the main qualitative case study for our pipeline : once available, we will run the trained model on these high-resolution tiles and compare the six-class coverage map against both the NParks mask and the original GMW labels.

This will give a detailed, location-specific view of where the refinement actually adds value–especially for narrow mangrove strips along tidal channels–and whether the Grad-CAM activations match ecological features visible at 1 m resolution.

Combining these images with field measurements could further extend the pipeline to estimate above- and below-ground carbon stocks, which is still the main gap for credible blue-carbon accounting (Rani et al. [16]; Ouyang et al. [17]).

Scaling to global coverage will need multi-GPU training.

Our single-GPU setup works well for regional studies but is too slow for full global deployment. Future work could also add uncertainty estimates through repeated inference with dropout enabled, which would flag the chips where the model is least confident and where expert review would be most useful.

V. Conclusion

We presented a pipeline that turns binary Global Mangrove Watch labels into six graduated coverage classes using AlphaEarth Foundations embeddings and a Vision Transformer.

The core contribution is a coverage ratio ρ_i (Eq. 2) computed by intersecting each chip with all overlapping GMW polygons, sorted by a spatial location index, then rounded into six classes (Eq. 3).

Non-mangrove chips ($\rho = 0$) are built by geographic random sampling with a spatial filter across four global regions, so the training data covers mangrove ecosystems worldwide and not just the most polygon-dense areas.

The ViT's self-attention lets every part of the chip communicate with every other part, capturing the spatial gradients that define mangrove boundaries.

Grad-CAM maps confirm that predictions focus on the right ecological features.

The pipeline needs no external ground truth, is fully reproducible, and is released as an open Python package.

Future work will add high-resolution validation imagery from Pulau Ubin and extend the approach toward carbon stock estimation.

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Note: *This paper presents ongoing research. Experimental results and quantitative evaluations are still being conducted. Final performance metrics will be reported upon completion. We aim at publishing a comprehensive version of this work in a peer-reviewed journal at the end of March.*

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